

Question. 10,000 miles –in Chicago - compare Hybrid, electric, ICE, calculate the emission saving

Answer:

Total mileage = 10,000

Gasoline price in Chicago (ICE – Hybrid) = \$3.5 per gallon

Electricity price in Chicago = \$0.16 per kWhr

ICE Efficiency = 25 MPG

Hybrid Efficiency = 40 MPG

Electric cars Efficiency = 3.4 mile/kWhr

Energy consumption = Total mileage / efficiency

ICE = $10,000 / 25 = 400$ gallon

Hybrid = $10,000 / 40 = 250$ gallon

Electric car = $10,000 / 3.4 = 2,941.1$ kWhr

Energy cost = Energy consumption × Price (gasoline/electricity)

ICE = $400 \times 3.5 = \$1400$

Hybrid = $250 \times 3.5 = \$875$

Electric car = $2,941.1 \times 0.16 = \$470.576$

Hybrid and Electric car saving = Hybrid/Electric car energy cost - ICE energy cost

1. Hybrid car Cost Savings = $1400 - 875 = \$525$
2. Electric car Cost Savings = $1400 - 470.576 = \$929.424$

The Electric Vehicle shows larger cost savings over 10,000 miles compared to the gasoline car (ICE).